PHYSICS 7B, Lecture 1 – Summer 2024
Midterm 2, C. Bordel
Tuesday, July 23rd, 11:10 am-12:30 pm

- Student name:
- Student ID #:
- Section #:

- ***Rules:*** *You may work on this exam from 11:10am-12:30 pm PT. This midterm is closed book and closed notes and you are **not** allowed to use any electronic devices, with the exception of a calculator. You are **not** allowed to communicate with any other students or discuss the content of the exam with anyone besides the Physics 7B teaching staff. Any violations of this policy will be considered a breach of academic integrity and result in disciplinary actions.*

- ***Equation sheet:*** *Only the equations given on the equation sheet can be used without derivation.*

- ***Partial/Full Credit:*** *We will give partial credit on this midterm, so if you are not sure how to do a problem, or if you do not have time to complete a problem, try to demonstrate your understanding of the physics involved by drawing a clear diagram and explaining (in physics terms) why you believe your answer to be incorrect, or how you would do the problem if you had time. Please be aware, however, that writing irrelevant or incorrect information will result in less partial credit. Remember that you need to show your work and justify your answers in order to get full credit.*

- ***Honors Statement:*** *It is expected that during this examination, as with any examination that they undertake at the university, students adhere to the usual standards of academic integrity at the University of California at Berkeley as outlined on by the Center for Teaching and Learning. Therefore, by submitting your exam, you are affirming the following statement:*
"I swear on my honor that I have neither given nor received aid on this exam. In addition, I abided by all of the examination policies as outlined above."

<u>Problem 1</u> – 25 points

A charge distribution is designed so that it forms a smiley face. An outer ring of radius $2R$ carries positive charge Q. The center of the ring is the origin of the (x,y) coordinate system. Two point-charges carrying charge Q are placed at distance R from the origin, and are symmetrically located at 45º from the y-axis. A half-ring of radius R, carrying charge Q, is placed symmetrically about the y-axis below the origin (Fig.1). All charge distributions are uniform. An electron of mass m and charge ($-e$) is positioned at the center of the static charge distribution. *The acceleration due to gravity has magnitude g.*

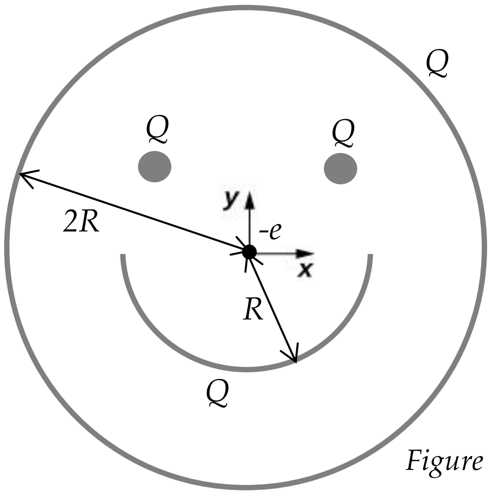

Figure 1

a. Determine, with or without calculations, the force exerted by the outer ring on the electron.

b. Determine the force exerted by the truncated ring on the electron.

c. Determine the force exerted by the two point-charges on the electron.

d. Could the electron – of mass m – remain at rest at the origin if the smiley face lies in a vertical plane? If so, determine charge Q. *Note that* $\pi > \sqrt{2}$.

Problem 2 – 25 points

A non-conducting solid sphere of radius R_1 is charged with a positive non-uniform charge density $\rho(r) = \frac{\rho_0 R_1}{r}$, where ρ_0 is a constant and r is the radial distance measured from the center of the sphere.

a. Determine the magnitude and direction of the electric field produced by the charged sphere at any point, both inside and outside the sphere. *Explain your choice of method to calculate the electric field.*

Initials:

b. Determine the electric potential at any point ($r \le R_1$ and $r \ge R_1$), assuming that *$V(0)=0$*.

A conducting spherical shell extending from R_1 to R_2 [gray shade] (Fig.2) is added around the previous non-conducting sphere [hatching lines] and the electric field is measured to be zero outside of the combined system *(r >* R_2*)*.

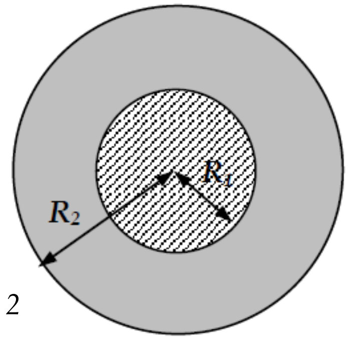

Figure 2

c. Determine the net electric charge Q_c carried by the conducting spherical shell.

d. Determine the surface and volume charge densities for the conducting spherical shell. Let σ_1 be the surface charge density on the inner surface, σ_2 the surface charge density on the outer surface, and ρ_c the volumetric charge density inside the conducting spherical shell. *Make sure you articulate your reasoning very clearly.*

<u>Problem 3</u> - 25 points

A thin wire carrying a uniformly distributed charge Q ($Q>0$) is bent into a semicircle of radius R. A rod of length R carrying the same charge Q, also uniformly distributed, is attached to the semicircle on the left, as shown in Figure 3. The x-axis points as shown in the figure and the origin is taken at the center of the semicircle.
An electron of mass m and charge $-e$ is released from rest at infinity on the x-axis.
Set $V=0$ at infinity on the x-axis.

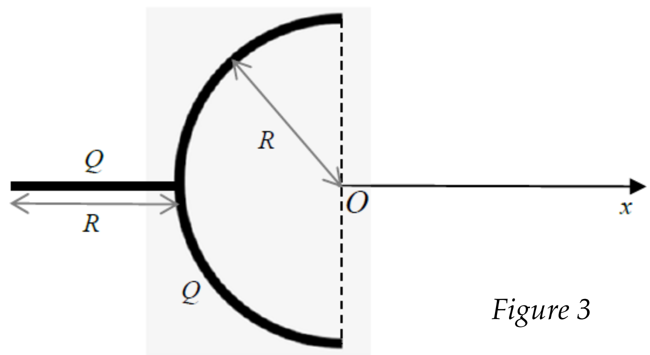

Figure 3

a. Determine the electric potential created by the semicircle at the origin.

b. Determine the electric potential created by the rod at the origin.

Initials:

c. Determine the change in electric potential energy of the electron from infinity to the origin along the x-axis.

d. Determine the speed of the electron as it reaches the origin, assuming it is released from rest at infinity on the x-axis.

Initials:

<u>Problem 4</u> – 25 points

Two very long cylindrical conducting shells of negligible thickness and radii R_1 and R_2 (with $R_2 > R_1$) are used to form a capacitor, with vacuum in the gap. Even though both shells carry electric charge on their surface, it is more convenient to define β, the electric charge per unit length of the cylinders. The shells carry equal-opposite charge distribution β. The inner shell carries a positive charge and the outer shell a negative charge.

a. Determine the magnitude and direction of the electric field created between the two shells.

b. Calculate the capacitance per unit length of the cylindrical capacitor.

Initials:

A network of capacitors is shown in Fig.4. All the capacitors are identical and have the same capacitance C.

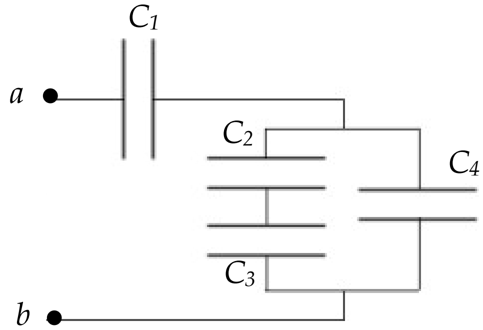

Figure 4

c. Determine the equivalent capacitance of the network of capacitors.

d. Determine the energy stored in this array of capacitors if a voltage V is applied between a and b.

Initials:

Electrostatics

$$\vec{F} = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2}\,\hat{r}$$

$$\vec{F} = Q\vec{E}$$

$$d\vec{E} = \frac{dQ}{4\pi\epsilon_0 r^2}\hat{r}$$

$$\lambda = \frac{dQ}{dl} \;\;;\;\; \sigma = \frac{dQ}{dA} \;\;;\;\; \rho = \frac{dQ}{dV}$$

$$\vec{p} = Q\vec{d}$$

$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$U = -\vec{p} \cdot \vec{E}$$

$$\Phi_E = \int \vec{E} \cdot d\vec{A}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$$

$$\Delta U = q\Delta V$$

$$V(b) - V(a) = -\int_a^b \vec{E} \cdot d\vec{l}$$

$$dV = \frac{dQ}{4\pi\epsilon_0 r}$$

$$\vec{E} = -\vec{\nabla}V$$

$$\vec{F} = -\vec{\nabla}U$$

Capacitors

$$Q = CV$$

$$C_{eq} = C_1 + C_2 \text{ (In parallel)}$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} \text{ (In series)}$$

$$\epsilon = K\epsilon_0$$

$$U = \frac{Q^2}{2C}$$

$$u = \frac{\epsilon}{2}\left|\vec{E}\right|^2 \text{ (energy density)}$$

Gradient and displacement

- ***Cylindrical coordinates :***

$$\vec{\nabla}f = \frac{\partial f}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial f}{\partial \theta}\hat{\theta} + \frac{\partial f}{\partial z}\hat{z}$$

$$d\vec{r} = dr\,\hat{r} + rd\theta\,\hat{\theta} + dz\,\hat{z}$$

- ***Spherical coordinates :***

$$\vec{\nabla}f = \frac{\partial f}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial f}{\partial \theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}\hat{\phi}$$

$$d\vec{r} = dr\,\hat{r} + rd\theta\,\hat{\theta} + r\sin\theta\,d\phi\,\hat{\phi}$$

Lengths, Areas and Volumes

Arc length: $dL = Rd\theta$

Cylinder: $A = 2\pi RH \;;\; V = \pi R^2 H$

Cylindrical shell: $dV = 2\pi RHdR$

Thin Circular Ribbon : $dA = 2\pi RdR$

Section of Thin Ribbon : $dA = RdRd\theta$

Sphere: $A = 4\pi R^2 \,;\, V = \frac{4}{3}\pi R^3$

Spherical shell: $dV = 4\pi R^2 dR$

Motion and Energy

$$\Delta K + \Delta U = W_{nc}$$

$$K = \frac{1}{2}mv^2$$

$$\overrightarrow{F_{sp}} = -k\,\vec{x} \,;\, U_{sp} = \frac{1}{2}kx^2$$

$$\frac{d^2x}{dt^2} + \omega^2 x = 0 \text{ (simple harmonic motion)}$$

Taylor Expansions near x=0

$$\sin(x) \approx x$$

$$\cos(x) \approx 1 - \frac{x^2}{2}$$

$$e^x \approx 1 + x + \frac{x^2}{2}$$

$$(1+x)^\alpha \approx 1 + \alpha x + \frac{(\alpha - 1)\alpha}{2}x^2$$

$$\ln(1+x) \approx x - \frac{x^2}{2}$$

Initials:

Integrals

$$\int_0^\infty x^n e^{-ax} dx = \frac{n!}{a^{n+1}}$$

$$\int_0^\infty x^{2n} e^{-ax^2} dx = \frac{(2n)!}{n!\, 2^{2n+1}} \sqrt{\frac{\pi}{a^{2n+1}}}$$

$$\int_0^\infty x^{2n+1} e^{-ax^2} dx = \frac{n!}{2a^{n+1}}$$

$$\int (1+x^2)^{-1}\, dx = \arctan{(x)}$$

$$\int (1+x^2)^{-\frac{1}{2}}\, dx = \mathrm{Ln}\,(\sqrt{1+x^2}+x)$$

$$\int x(1+x^2)^{-\frac{1}{2}}\, dx = \sqrt{1+x^2}$$

$$\int (1+x^2)^{-\frac{3}{2}}\, dx = \frac{\mathrm{x}}{\sqrt{1+x^2}}$$

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$\int \frac{x}{1+x^2} dx = \frac{1}{2}\ln{(1+x^2)}$$

$$\int \frac{1}{x} dx = \ln{(x)}$$

$$\int \frac{1}{\cos(x)} dx = \ln\left(\left|tan\left(\frac{x}{2}+\frac{\pi}{4}\right)\right|\right)$$

$$\int \frac{1}{\sin{(x)}} dx = \ln\left(\left|tan\left(\frac{x}{2}\right)\right|\right)$$

Trigonometry

$$\sin(2x) = 2\sin(x)\cos{(x)}$$

$$\cos(2x) = 2cos^2(x) - 1$$

$$\sin(a+b) = \sin(a)\, scos(b) + \cos(a)\sin{(b)}$$

$$\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin{(b)}$$

$$1 + \cot^2(x) = \csc^2(x)$$

$$1 + \tan^2(x) = \sec^2(x)$$

$$c^2 = a^2 + b^2 - 2ab\cos(\theta) \quad \text{(law of cosine)}$$